

Shervin Naseri

Senior Software Engineer | Distributed Systems & Python Specialist

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PROFESSIONAL SUMMARY

High-performance Software Engineer with 10 years of experience designing scalable distributed systems, event-driven architectures, and secure backend applications. Proven track record re-architecting core software components to interface seamlessly with cloud-based AI services, relational SQL, and NoSQL databases. Expert in object-oriented design and implementing robust security governance and DevOps workflows.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C++, SQL, Batch, HTML, CSS
Application Architecture: Object-Oriented Design, REST Services, Distributed Systems, Event/Message-Driven Architectures (ZeroMQ, BullMQ)
Data & Cloud Storage: SQL (PostgreSQL, MySQL, SQLite), NoSQL (MongoDB, Redis, Neo4j, Weaviate), Amazon Web Services (AWS), Prisma ORM
DevOps & Security: CI/CD (GitHub Actions), Docker, Docker Compose, Docker Swarm, Nginx, Ansible, Role-Based Access Controls
Certifications: AWS Certified Generative AI - Professional

PROFESSIONAL EXPERIENCE

AI & Software Engineering Consultant

Independent Contracts | Ottawa, ON

March 2025 - Present

- Architected asynchronous Model Context Protocol (MCP) server backends using FastAPI and Python 3.12, utilizing SQLAlchemy with asyncpg for high-throughput PostgreSQL interactions.
- Engineered event-driven file monitoring systems leveraging asyncio and message-driven task queues with Redis and BullMQ, implementing robust state tracking and concurrency mechanics.
- Implemented advanced Python decorator patterns to enforce role-based privilege escalation, standardizing cross-module error encapsulation and secure development practices.

Senior ML Engineer

Pardazesh Etelaat Kian (IraPardaz) | Tehran, Iran

June 2020 - Sep 2021

- Designed and deployed high-concurrency, multiprocessing data engineering pipelines, managing 2GB payload chunking and executing low-level system hooks to guarantee continuous uptime.
- Re-architected core software components using modern object-oriented design patterns, improving system availability, scalability, and structural resiliency across complex layouts.
- Collaborated with cross-functional engineering and AI teams to lead technical hiring processes, ensuring alignment with enterprise architecture standards and internal project governance.

Core Backend Engineer

Nikaat Corp | Tehran, Iran

Dec 2016 - April 2020

- Architected, developed, and maintained distributed web services and REST APIs using Node.js and modern application architectures, ensuring high availability for multi-tenant portfolios.
- Built and maintained data storage solutions across relational systems and NoSQL instances (MongoDB), implementing thread-safe locking and asynchronous data persistence.
- Managed end-to-end DevOps workflows, containerizing application clusters via Docker and Docker Swarm while establishing automated CI/CD pipelines through Nginx proxies.

Python Developer

Vestaak | Tehran, Iran

March 2016 - Dec 2016

- Developed personalized recommendation engines and RESTful APIs using Python, processing large-scale unstructured messaging data mined via external API integrations.

SELECTED PROJECTS

CodebaseMCP (Asynchronous Analysis Server) | [Github](#)

- Architected an API-driven server using FastMCP and Python, exposing interactive tools for external systems to semantically search, inspect, and isolate environments across multi-tenant vector databases.
- Engineered an event-driven background processing engine with asyncio and debounce mechanics to synchronize real-time file system monitoring with local data backends without blocking main event loops.

GameASR (Distributed Messaging Bridge) | [Github](#)

- Engineered a polyglot, language-agnostic Remote Procedure Call (RPC) bridging system using ZeroMQ, establishing high-throughput messaging and async inter-process communication between distributed clients.
- Implemented automated schema tool-calling, context managers, and multi-hop graph traversals natively within a decoupled architecture, optimizing token usage and handling high-concurrency execution loops.

EDUCATION

MSc. Computer Science

University of Ottawa | Ottawa, ON

Sep 2021 - March 2025

- Research Focus: Simulation Reliability in Autonomous Systems. Co-authored technical research on testing autonomous systems published in Empirical Software Engineering (EMSE 2024).

MSc. Software Engineering

Sharif University of Technology | Tehran, Iran

Graduated

- Curriculum: Advanced Software Architecture, Algorithm Design, and Database Management. Thesis focused on Large-Scale Network Similarity and predictive modeling architectures using BERT.

BSc. Computer Science

Amirkabir University of Technology | Tehran, Iran

Graduated

- Focus on Data Structures, Core Algorithms, and Information Retrieval. Published pure algorithmic research in ICCG 2019.